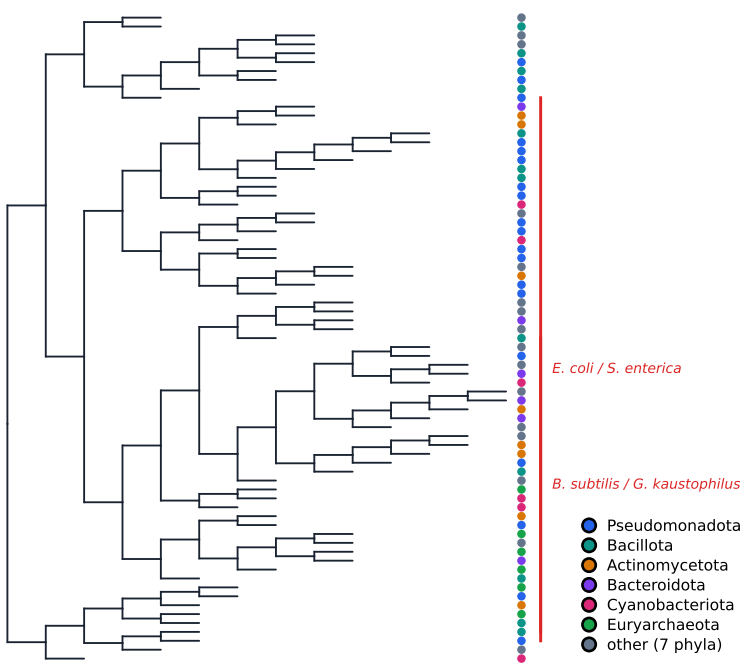


Figure 6. Real-data validation

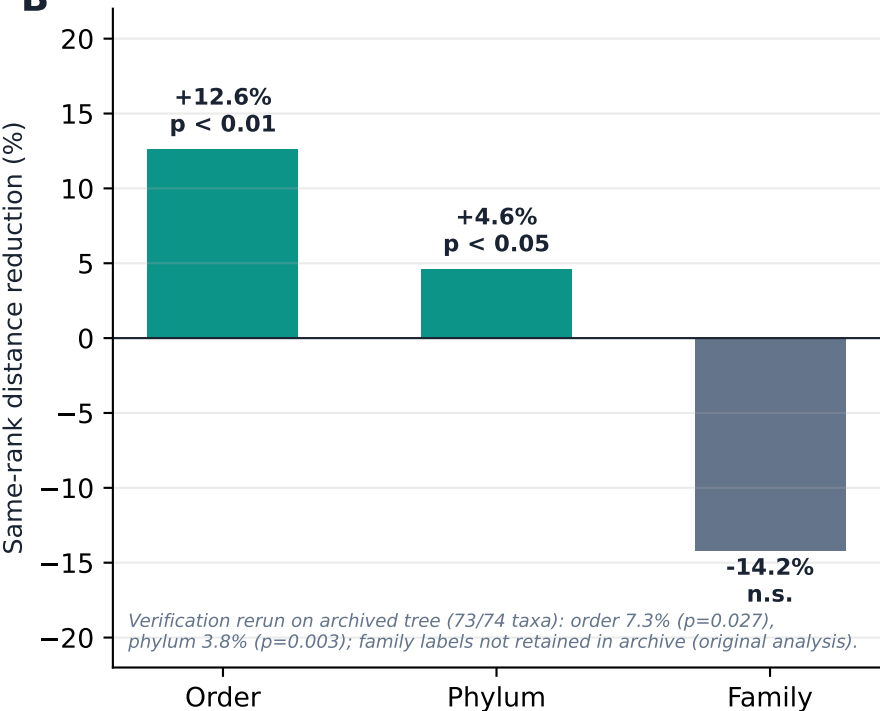
A

16S rRNA tree, 73 taxa (k=5,gap2; 1.2 s, no alignment)



B

Phylogenetic signal by taxonomic rank



C

Fusang v1 (2023) vs Tardigrade Edition

	Fusang v1	Tardigrade
Representation	neural net features	k-mer frequency vectors
Scalability	4-40 taxa	10,000+ taxa
Speed	GPU-dependent	CPU-only; 10k taxa < 1 min
Deployment	Docker / cloud	single script + bundled binary
Indel handling	not modelled	spaced k-mer sampling

D

Gene-length DNA benchmark (13 mtDNA loci, 25 fish)

